

MODULE 2 · THE AUTOMATION LADDER

RPA: The Tactical Integration Layer

The first rung of automation — automate the rule, route the exception

Friday, September 25, 2026 · 8:00–10:30 AM · 2 MTC Room 802

IE-GY 9113B — Systems Integration: From ERP to Agentic AI · Prof. Emanuele Cimica

Today: activate the note, work two cases, then build your scan

1 Activation 8:05

Rebuild the week's spine from memory — the floor everyone joins from.

2 Reconnect + story 8:19

Back to obliterate-then-automate and your Lab 1 decision point; then Thermo Fisher.

3 Two cases, worked 8:35

Automate the rule, route the exception — Thermo Fisher and Bailey, then the scan on Citi Bike.

4 Apply — 3 breakouts 9:03

By team, you finish your Lab 1 map and build the \$4 piece: your RPA-candidate scan.

5 Prime Week 5 10:13

Scaling RPA and the Center of Excellence — why bots stall past a handful.

6 Homework + wrap 10:22

The RPA-candidate scan · read the Week 5 note.

Q1 — Where are we on the ladder?

make agentic

AI agents (Module 3) → the agentic enterprise (Module 4)

make intelligent

Intelligent automation (Module 3)

automate

RPA — this week (Module 2)

◀ you are here

integrate

The integration backbone — one single source of truth (Module 1)

Q2 — What is a “bot,” really?

Works the screens

Not the plumbing.

RPA drives a company's existing applications through the user interface — clicking, reading fields, copying between screens by a fixed script. A robot that does the human's cross-system carrying.

Fast, and brittle

Speed traded for robustness.

It bridges un-integrated systems in days, not months — but it breaks the moment a screen it depends on changes. A tactical patch over a missing integration, not a fix.

Surface, not structure. RPA automates the symptom of a missing foundation — a person moving data between systems. Powerful and quick; inherently fragile.

Q3 — Three ways to take a human out of the loop

Traditional

In the plumbing.

APIs, database, native integration — beneath the screen. Robust, but needs engineering to build and change.

Bailey's NetSuite O2C.

RPA

At the surface.

The same screens a person uses. Rules-based, fast to build, brittle when a screen moves.

Thermo Fisher's order-entry bot.

AI

On judgement.

Unstructured, ambiguous input — no fixed rule. Reads intent, classifies, resolves.

The exception (previewed; Week 7).

Rule vs. judgement is the line. It's the boundary you drew by hand in Lab 1 — and it organizes the whole module.

Q4 — What makes a good RPA candidate?

Repetitive

High, recurring volume.

The same work many times, so a bot's fixed cost pays back.

Rules-based

Complete, explicit logic.

Every case decided by a rule — no point that needs judgement.

Stable

Screens rarely change.

A moving screen breaks the bot's brittle, surface-level script.

All three, or it isn't a candidate. The step that passes all three is your Lab 1 stable-rules decision point — and your Lab 2 (Make) target.

Q5 — When should a bot NOT be built?

The task mindset

Bolt a bot on a tedious step.

A few hours come back — while the tangled process it lives in stays tangled. A faster broken process.

The process mindset

Ask what the whole flow is for.

Redesign first; automate the stable residue that survives. Obliterate, then automate — Week 2's Ford.

Automate a broken process and you pave the cow path. RPA is the “then automate” half — applied after redesign, never instead of it. Process first, bot last.

Q6 — Where is all this repetitive work made?

MRP + S&OP

The batch-push planning layer.

MRP explodes a plan into component orders (Orlicky, 1975);
S&OP reconciles demand with supply (Ling & Goddard, 1988).

The rhythm

Overnight → morning firefighting.

Plans compute overnight, cascade downstream; humans clear the exceptions at dawn — a PO to release, a mismatch to reconcile.

RPA automates the firefighting, not the fire. It makes the batch-push symptoms cheaper. The deeper cure — event-driven operations — is later work (Modules 3–4).

The spine, in one line

A working backbone throws off repetitive, rules-based exception work. Sort it: automate the stable-rules step, hand judgement to AI, send the broken step back to redesign.

- Repetitive + rules-based + stable → RPA candidate → your Lab 2 (Make) build.
- Needs judgement → the AI-agent target → Lab 3 (Relevance AI, Week 9).
- Changes constantly / broken → redesign first (obliterate, then automate — Week 2).

This week you sort the work. You are not building yet — you are producing the RPA-candidate scan that becomes Board Memo §4.

You already found this week's target

From Week 2

Obliterate, then automate. Ford eliminated the invoice-mismatch work — it didn't automate the chasing of mismatches faster.

From Lab 1

You marked three diagnostic points on your process — two of them decision points: one with stable rules, one that needs judgement.

This week

The stable-rules decision point IS your RPA candidate. Today you formalize it and justify it — the scan for §4.

The room already did the diagnosis. Week 4 turns the point you labelled by hand into a board-ready RPA-candidate scan.

Thermo Fisher — the human was the integration

The company

One of the world's largest suppliers of scientific instruments — years of mergers, a sprawl of disparate ERPs and CRMs. 10M+ transactions and interactions a year.

The pain

A rep answering “where is my order?” navigated nine systems and nine screens; a routine call ran past ten minutes. A person carrying data between systems that couldn't talk.

Through the lenses

Frustration (staff and customers) and Time (ten minutes for a one-line answer) — Operations first, with an immediate Customer face.

Olavsrud, CIO, 12 Aug 2022 — verified figures

Project Northstar: automate the rule, route the exception

The build

RPA + machine learning + OCR ingest incoming orders — including faxed and emailed ones — and move the data across the systems of record automatically. No months-long ERP consolidation.

The split

The repetitive, rules-based order entry is automated; the ambiguous order exception is routed out to a case-management portal — to where judgement lives.

Scoped first

Processes were mapped to the click level with a delivery partner before a line was automated. Process-first, bot-last — which is why Lab 1 came before this week.

The RPA-candidate filter, in production. Automate what has a rule; hand the exception to a human. Same move you'll make on your own map.

A dull step, done at scale — on two threads

-27%

average call hold times, after full deployment — a direct Customer gain.

Cash

faster order processing accelerated cash flow — a direct Cash gain.

The win came from a bot doing something dull, reliably, at volume. Not from cleverness — from a step that was repetitive, rules-based, and stable, while the step that was none of those went to a human.

One move, reusable on any process

“Automate the rule, route the exception” is the whole craft of the week. Run every step of a mapped process through three tests and it sorts into three destinations.

Passes all three

→ **RPA**

Repetitive + rules-based + stable.
Your Lab 2 (Make) build.

Needs judgement

→ **AI**

Two reasonable people route it differently. Lab 3 (Week 9).

Broken / unstable

→ **Redesign**

Automating it just paves the cow path. Fix it first.

The scan is this sorting, written down. Two or three real candidates — and an honest account of what you rejected, and why.

Which tool for which step of an order-to-cash flow?

Traditional

Order → inventory reservation → ledger entry, underneath the interface. What Bailey has: native integration, no screen involved.

RPA

A bot logs into the order tool, reads the fields, types them into the separate finance system — same outcome, riding the surface, brittle to a screen change.

AI

The order that arrives as a free-text email, or the dispute that fits no rule — where something must interpret, not transcribe.

Same flow, three tools. Chosen by what each step actually requires — not by which is newest.

Bailey Hydraulics — what “working” looks like first

Before

A mid-market hydraulic-components manufacturer on a customized legacy ERP, storefront bolted on separately — orders moved by hand between web and back end. Close: 20 days.

After

One NetSuite platform gave order-to-cash a single source of truth; placement and fulfilment became continuous. Financial close fell from 20 days to 3.

The lesson

That gain came from native integration — the plumbing — not a bot. When systems become one, whole “carry data between screens” tasks simply vanish.

Ask what integration already owns — before you point a bot at a seam. Given the backbone works, what would RPA still change? Often: less than you'd think, at the manual edges.

Why a good candidate is worth a bot

≈0

marginal cost per transaction after build — variable labour becomes a fixed cost.

A worked estimate

3 minutes × 500 times a day ≈ 25 hours a day — three to four people.

A bot takes out the rules-based bulk (exceptions still go to a human). Board case = recurring labour removed vs. one-time build + licence, as a payback period.

The business case sharpens with volume. Which is why “repetitive” is the first test — a bot on low volume is a fixed cost with nothing to spread it over.

The same sentence describes a person

“The marginal cost collapses toward zero” means the hours a human used to spend are no longer needed for that task. RPA changes — and sometimes eliminates — back-office roles.

Sometimes real

Freed from moving data,
a specialist works the judgement-heavy cases a bot can't touch — genuinely better work.

Sometimes a story

“Redeployment to higher-value work”
can be the phrase a firm uses to avoid a harder conversation.

Ethics thread — hold the question open. When RPA removes the routine part of a role, what does the firm owe the person? Board Memo §7. Most firms Deloitte surveyed hadn't even calculated the workforce impact.

“Rules-based” does not mean “no human needed”

Attended

Runs beside a person.

Triggered by them, on their machine — a hand on the wheel, ready to take over. For work near judgement.

Unattended

Runs on its own.

On a schedule or a queue, no human present — clearing routine volume overnight. For work that needs no case-by-case eyes.

A bot follows a wrong rule with perfect confidence. So route edge cases, high-value transactions, and irreversible actions to a human checkpoint. You'll name that oversight in Lab 2; Week 5 governs it.

Settled practice, not a bet — Deloitte (2022)

74%

already implementing RPA — it is infrastructure, not an experiment.

Payback: 16 → 22 mo

Longer, not worse — as firms invest more to deliver more. The >70% cost cut one exec reported came only after radically re-engineering.

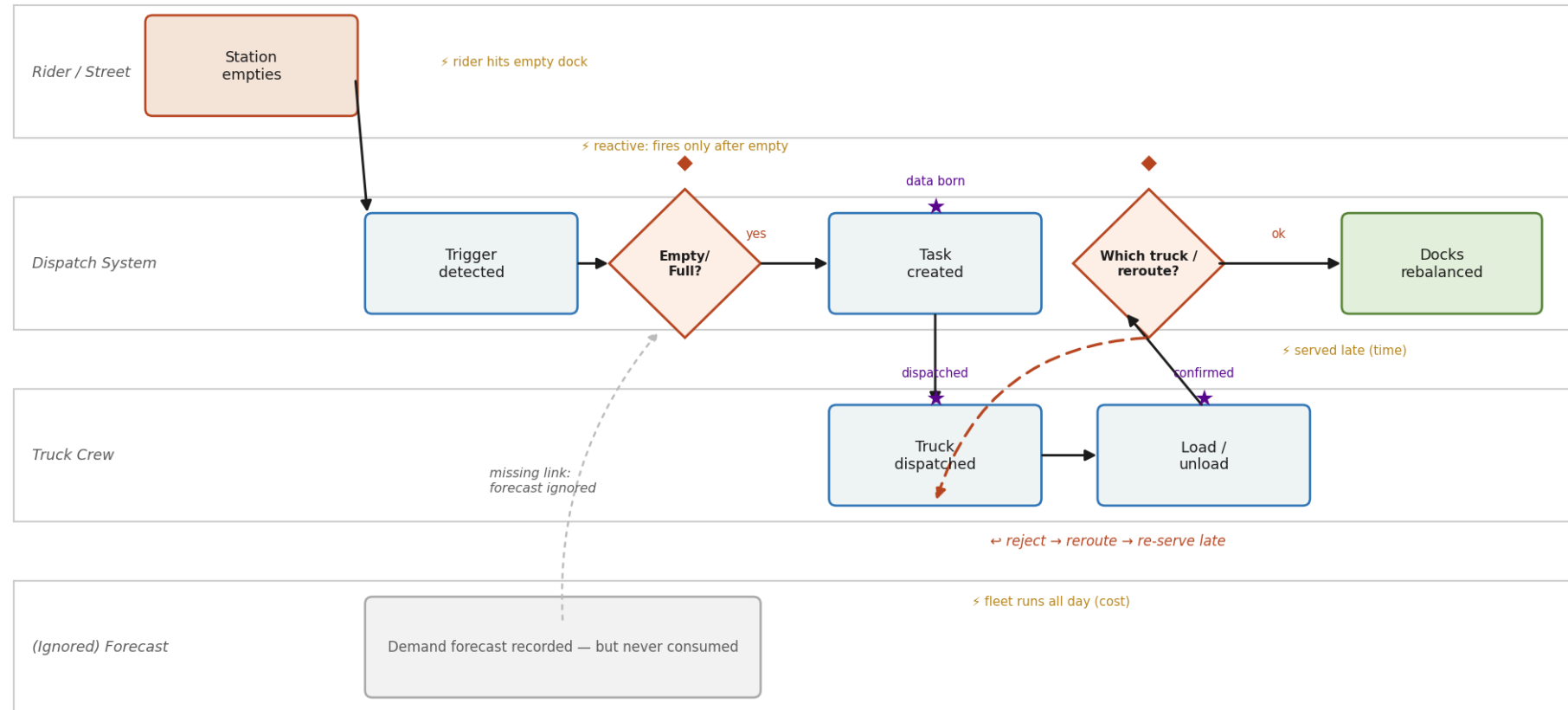
Barriers

Process fragmentation, unclear vision, IT readiness, resistance to change — none of them the technology. That's Week 5's Center of Excellence.

RPA is where you start, not where you stop. ~44% of mature adopters were already moving from task bots toward end-to-end automation, another ~48% planning to. Cite every figure to Deloitte (2022).

Citi Bike — you built this map in Lab 1

Citi Bike rebalancing — Order-to-Cash-equivalent flow (reactive) · IE-GY 9113B Lab 1 worked example



★ control point — data is born ♦ decision point — routing choice - - - rejection / rework path ⚡ value leak (Four Lenses)

Two decision points ♦ are today's whole question — one is a rule, one is a judgement. Synthetic worked example — a mirror, not a template; all figures synthetic.

The scan, worked — two points, two destinations

◆ Empty/Full? → RPA

Repetitive · rules-based · stable.

Automate the trigger → Lab 2 (Make). Brittle where it rides the nightly status feed — so build the human checkpoint, then break it on the stale feed. Thread: Operations (empty docks → Customer; fleet hours → Cash).

◆ Which truck / reroute? → AI

Two stations compete — no clean rule.

The call a rule can't make: which shortfall to serve first when trucks are scarce. Judgement → Lab 3 (Relevance AI, Week 9). Not this week's target.

RPA automates the firefighting, not the fire. The bot speeds the reactive trigger; the prize — the demand forecast the system already records but never consumes — is a redesign for later. Synthetic: ≈ 1 in 7 of 57 rebalancing tasks hit rework, re-served 40–70 min late. This is the shape of your \$4 scan → slide S6.

APPLY · BY PROJECT TEAM

Build your RPA-candidate scan

Three breakouts on your own Lab 1 map — sort the work, run the three tests, then cost and govern it. The output is Board Memo §4 (Automation opportunity, slide S6).

Finish your map, then sort the work

Pull up your Lab 1 map. Confirm it carries your stable-rules point and your judgement point — finish any gap — then sort its steps by which tool each needs.

FLOOR · start here

- Open your Lab 1 map: confirm your stable-rules ♦ point and your judgement ♦ point are on it. Fill any gap now — flag me if you're stuck.
- Walk it end to end; tag each step traditional / RPA / AI.
- Circle the one or two steps a bot could plausibly run.

STRETCH · if you read

- Name what native integration already handles — so a bot isn't pointed at a seam a real integration should own.
- Flag any step that only looks rules-based until you inspect it (as Citi Bike's Empty/Full trigger is — until the stale feed).

Output A finished map + a tagged process — each step marked traditional / RPA / AI.

Time
20 min
+ 3 debrief

Debrief: two teams name their clearest RPA step and one they were tempted to mis-tag.

Citi Bike, tagged — and the tempting mis-tag

Traditional

Consume the demand forecast the system already records but ignores — an integration / event-wiring fix, not a bot. The map's missing link.

RPA

The Empty/Full? trigger ♦ — detect the shortfall, fire the dispatch. Stable rules, while the live feed holds.

AI

Which truck / reroute? ♦ — the call when two stations compete for one truck. Judgement, not a rule.

Look past the label; test the real logic. “Which truck?” tempts a rule — “send the nearest” — until trucks are scarce; and the Empty/Full rule is only “stable” until it rides the stale nightly feed. Same step, RPA at one company, AI at another.

Run the three tests

Run the candidate filter across your process and produce the scan itself — the piece that becomes Board Memo §4. Teams present.

FLOOR · start here

- For each RPA-tagged step, test: repetitive? rules-based? stable?
- Keep the two or three that pass all three.
- For each, record: the step, why it passes, and the thread it moves.

STRETCH · if you read

- Name the prize per candidate — the board-level consequence (days of cash, fewer rejects).
- Write the honest rejections: one “redesign first,” one “needs judgement.”

Output The RPA-candidate scan: 2–3 candidates + threads + prize, and what you rejected.

Time
20 min
+ 6 present

Debrief: each team presents its scan — the §4 deliverable — for 2 minutes.

Citi Bike's scan, written as §4

Candidate

Empty/Full? trigger → RPA (Make, Lab 2). Thread: Operations (empty docks → Customer; fleet hours → Cash). Prize: fewer late dispatches, more shortfalls closed at lower cost.

Redesign first

The reactive structure itself. Consuming the forecast the system already records is the real prize — but a redesign, not a bot.

Needs judgement

Which truck / reroute when stations compete → an AI target (Lab 3), not RPA.

One honest candidate, its thread and prize, and what you refused. That's a §4 entry a board can read — the Automation-opportunity slide (S6), still inside Diagnose. The scan is the argument; the build is Lab 2.

Cost it, and govern it

Take one candidate from your scan and make the board case — the number, and the human safeguard.

FLOOR · start here

- Estimate the volume $\times$ time saved for one candidate.
- Sketch the simple payback: labour removed vs. build + licence.
- Say whether the bot is attended or unattended.

STRETCH · if you read

- Name where a human stays in the loop — which edge cases or irreversible actions get a checkpoint.
- State the workforce obligation the firm takes on if the role changes.

Output A one-candidate ROI sketch + the human-oversight and workforce note.

Time
16 min
+ 5 assemble

Debrief: two teams give their number and the single human checkpoint they'd keep.

What leaves the room · Citi Bike, costed and governed

Your scan

2–3 candidates with thread and prize; the honest rejections. → Board Memo §4, slide S6 (Automation opportunity), still Diagnose.

Citi Bike · cost

The Empty/Full bot trades recurring dispatch-monitoring time for a one-time build — order-of-magnitude, not a promise; watch-the-screen time moves to the judgement calls.

Citi Bike · human

Unattended on the live feed — but held to a human when the feed is stale or a dispatch is already committed. The stale nightly feed is where it breaks.

You've named where a bot belongs — and where a human stays. In Lab 2 (Week 6) you build the Empty/Full candidate in Make, then break it on the stale feed. The scan is the argument; the build is next.

Next: why bots stall past a handful

The problem

Scaling RPA.

The brittleness you met today, multiplied: screens move, selectors break, integrations drift. Most programmes stall at a handful of bots.

The answer

The Center of Excellence.

The governance and operating model that lets automation scale past the pilot — standards, ownership, and a human-in-the-loop by design.

The hardest barriers are organizational, not technical. Read the Week 5 note before the next session — it answers the scaling question you'll now start to feel.

Study question to bring

What is the single biggest governance risk of deploying RPA without a Center of Excellence?

Answer from what you know now — then note what you expect Week 5 to add. Think about the bot that follows a wrong rule, at scale, with no one watching.

Where should a human stay in the loop on purpose? Even when the rules would let the bot run alone — the question that grows louder every week of this course.

Where this points

- §4 done today → §5 (Agent Design) is Module 3; §7 (Human-AI Partnership) carries the workforce question.
- Your stable-rules candidate → Lab 2 (Make, Week 6): build it, then break it.
- Your judgement step → intelligent automation (Week 7) and the agent build (Lab 3, Week 9).

One robot, three threads. Thread-tag every candidate you find — a board funds a thread outcome (Operations, Cash, Customer), not a bot.

Discussion Forum 1 launches today — on Brightspace

What & when

Module 1 · Brightspace → Discussions · 10% of grade.

Post your response by Wed Sep 30. Two peer replies by Sat Oct 3.

How it works

A 300–350-word post, then two replies.

Graded on depth of analysis — not length, not agreement.
Substantive = challenge an assumption with evidence, add a case or data point, or offer a reasoned alternative. Not “I agree” or a summary.

Why it's here. It turns your Lab 1 diagnosis into a written argument — and rehearses the reasoning your Board Memo §2–§3 will need. Disagree with ideas; respect people.

Discussion Forum 1 — the prompt

The leak

Building on your Lab 1 diagnosis — not a new case — name the single leak where your process most fails to leverage. Say which of the Four Lenses surfaces it and which thread it rings. Then make the call: obliterate and redesign, or adopt the platform standard and leave it alone? Justify the choice.

The claim

Then take a position on the module's central claim — that a failure to leverage is usually a management problem wearing a technology costume. Does your diagnosis bear that out, or did you hit a genuine technology constraint? Argue from what you found.

What good looks like. ~300–350 words: a claim, grounded in evidence from your Lab 1 work, and a defensible position. The full guidance lives on Brightspace.

Before next week

- 1 Homework** Finish your RPA-candidate scan → Board Memo §4 (Automation opportunity, slide S6).
- 2 Read** The Week 5 note — Scaling RPA + the Center of Excellence — before the next session.
- 3 Set up** Make account before Lab 2 (Week 6). AWS or custom path: notify and onboard now.
- 4 Discussion Forum 1** Post by Wed Sep 30, two replies by Sat Oct 3 — Brightspace → Discussions (10%).

The scan is the argument; the build is Lab 2. You leave today with where a bot belongs on your company's backbone — evidence a board can act on.

WEEK 4 · RPA

Automate the rule. Route the exception.

Next · Week 5: Scaling RPA + the Center of Excellence

Discussion Forum — grading rubric

Criterion (weight)	Unacceptable — 0%	Minimal — 70%	Competent — 85%	Effective — 90%	Mastery — 100%
Original Post: Analytical Argument 50%	A summary of readings or a generic opinion; no argument, no counterargument, and not actionable.	An argument is attempted but stays close to summary; reasoning is thin and no limitation or counter-view is acknowledged.	A clear analytical argument that goes beyond summary and reads as usable, though the counterargument or limitation is treated lightly.	A well-reasoned argument that acknowledges a real counterargument or limitation and gives an executive something to act on.	A crisp argument that advances a defensible position, engages the strongest objection, and lands as executive-ready analysis.
Original Post: Grounding in Cases & Frameworks 20%	No specific reference to a course case or framework.	A vague or incidental reference that does not shape the argument.	At least one specific, correct reference that supports the point.	References are chosen well and integrated into the reasoning.	Evidence is precise and connective, showing command of the module.
Substantive Peer Engagement (two responses) 30%	Responses missing, or replies that only agree, restate, or offer generic observations.	One substantive response, or two that only partly clear the bar (mostly affirmation with little new evidence).	Two responses that each add something — a case, a data point, or a reasoned alternative.	Two responses that challenge an assumption with evidence or offer a reasoned alternative that advances the thread.	Two responses that reshape the discussion, introducing evidence or a reframing the original author would have to reckon with.